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A Assumptions and Proofs

Assumption A.1 (Binary Tokens). For simplicity and without loss of generality, we assume that all datasets share a uniform token table containing only binary tokens (specifically $\emptyset/1$).

Theorem A.2 (Ideal Code Length). Consider a finite parameter model $p_\theta(\cdot)$ training on dataset $\mathcal{D}$, the ideal code length $\mathcal{L}_{p_\theta}(\mathbf{x})$ of a random response $\mathbf{x}$ compressed by p_θ can be expressed as follows:

$$\mathbb{E}[\mathcal{L}_{p_\theta}(\mathbf{x})] = \left\lceil \frac{|\mathbf{x}|}{d} \right\rceil \left[- \sum_{l=1}^d \sum_{j=1}^{2^{l-1}} p_{lj} \log p_{lj} \right],$$

where d represents the depth of the $\mathcal{T}_\mathcal{D}$ after pruning under Definition 3.3 protocol, p_{lj} represents the probability values of the EOS nodes for the j -th node at the l -th layer.

Proof. When $|\mathbf{x}| \leq d$, the compression protocol defined in Definition 3.3 can perfectly compress $\mathbf{x}$. Hence, the expectation of the ideal code length $\mathcal{L}_{p_\theta}(\mathbf{x})$ satisfies:

$$\mathbb{E}[\mathcal{L}_{p_\theta}(\mathbf{x})] = \left[- \sum_{l=1}^d \sum_{j=1}^{2^{l-1}} p_{lj} \log p_{lj} \right],$$

where d represents the depth of the pruned tree $\mathcal{T}'_\mathcal{D}$ and p_{lj} represents the probability values of the EOS nodes for the j -th node at the l -th layer.

Now consider $sd \leq |\mathbf{x}| \leq (s+1)d$. Let us suppose that $\mathbf{x} = (\mathbf{x}_1 \cdots \mathbf{x}_s \mathbf{x}_{s+1})$, where $|\mathbf{x}_k| = d$, for $k \in \{1, \dots, s\}$ and $|\mathbf{x}_{s+1}| \leq d$. In this case, $\mathbf{x}$ cannot be perfectly compressed by the model. Hence, the compression of $\mathbf{x}$ needs to be performed in segments, and the length of each segment is not greater than d .

$$\begin{aligned} \mathbb{E}_\mathbf{x}[\mathcal{L}_{p_\theta}(\mathbf{x})] &= \mathbb{E}_{\mathbf{x}_1}[\mathcal{L}_{p_\theta}(\mathbf{x}_1)] + \mathbb{E}_{\mathbf{x}_1} \mathbb{E}_{(\mathbf{x}_2 \dots \mathbf{x}_{s+1})}[\mathcal{L}_{p_\theta}((\mathbf{x}_2 \dots \mathbf{x}_{s+1})) | \mathbf{x}_1] \\ &= \mathbb{E}_{\mathbf{x}_1}[\mathcal{L}_{p_\theta}(\mathbf{x}_1)] + \sum_{\mathbf{x}_1} p(\mathbf{x}_1) \sum_{(\mathbf{x}_2 \dots \mathbf{x}_{s+1})} p(\mathbf{x}_2 \dots \mathbf{x}_{s+1} | \mathbf{x}_1) \cdot \mathcal{L}_{p_\theta}((\mathbf{x}_2 \dots \mathbf{x}_{s+1})) \\ &= \mathbb{E}_{\mathbf{x}_1}[\mathcal{L}_{p_\theta}(\mathbf{x}_1)] + \mathbb{E}_{(\mathbf{x}_2 \dots \mathbf{x}_{s+1})}[\mathcal{L}_{p_\theta}((\mathbf{x}_2 \dots \mathbf{x}_{s+1}))] \\ &= \sum_{k=1}^{s+1} \mathbb{E}_{\mathbf{x}_k}[\mathcal{L}_{p_\theta}(\mathbf{x}_k)] \\ &= \left\lceil \frac{|\mathbf{x}|}{d} \right\rceil \left[- \sum_{l=1}^d \sum_{j=1}^{2^{l-1}} p_{lj} \log p_{lj} \right], \end{aligned}$$

thus the proof is completed. $\square$

Definition A.3 (Mass Distribution in Token Tree). Consider the sample space Ω consisting of all responses in dataset $\mathcal{D}$. The probability distribution $\mathcal{P}_\mathcal{D}$ of all subtrees at the d -th level nodes of $\mathcal{T}_\mathcal{D}$ is a mapping from Ω to $[0, 1]$. Let $X_\mathcal{D}$ be the random variable representing the probability value taken at each leaf multiplied by the number of leaves. The mass distribution $\mathcal{P}_{mass}$ represents the probability that $X_\mathcal{D}$ takes the corresponding probability value. According to the definition of $\mathcal{P}_{mass}$, $\mathbb{E}[X_\mathcal{D}] = 1$.

Remark A.4 (Mixture of Mass Distribution). For independently and differently distributed datasets $\mathcal{D}_1, \dots, \mathcal{D}_N$, $\mathcal{D} = \bigcup_{i=1}^N \mathcal{D}_i$ is a mixture of these datasets. For the pruned trees $\mathcal{T}_1, \dots, \mathcal{T}_N$ of these datasets with depth d , the random variables of their leaf nodes satisfy the following relationship:

$$X_\mathcal{D} = \frac{\sum_{k=1}^N |\mathcal{D}_k| X_{\mathcal{D}_k}}{\sum_{k=1}^N |\mathcal{D}_k|},$$

where $X_{\mathcal{D}_k}$ follows the mass distribution $\mathcal{P}_{mass}^k$. For $X_{\mathcal{D}_{k_i}}$ and $X_{\mathcal{D}_{k_j}}$ from different datasets, $X_{\mathcal{D}_{k_i}}$ and $X_{\mathcal{D}_{k_j}}$ are independent of each other.

Lemma A.5 (Entropy of Mass Distribution). *Consider the pruned trees $\mathcal{T}'_k$ and $\mathcal{T}'$ of dataset $\mathcal{D}_k$ and $\mathcal{D} = \bigcup_{i=1}^N \mathcal{D}_i$. Denote that the response distribution and the mass distribution of $\mathcal{T}'$'s tree nodes are $\mathcal{P}^{\mathcal{D}}$ and $\mathcal{P}_{mass}$. Similarly, we define the response distribution $\mathcal{P}_k^{\mathcal{D}}$ and the mass distribution $\mathcal{P}_{mass}^k$. When the number of the leaf nodes M is sufficiently large, the cross-entropy of the response distribution can be rewritten as follows.*

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{P}_k} [-p^{\mathcal{D}_k} \log p^{\mathcal{D}}] = \mathbb{E}_{X_{\mathcal{D}_k} \sim \mathcal{P}_{mass}^k, X_{\mathcal{D}} \sim \mathcal{P}_{mass}} [-X_{\mathcal{D}_k} \log X_{\mathcal{D}}] + \log M,$$

where $p^{\mathcal{D}}, p^{\mathcal{D}_k}$ stand for the probability of the leaf nodes of $\mathcal{T}', \mathcal{T}'_k$ while $X_{\mathcal{D}_k}, X_{\mathcal{D}}$ stand for the random variables of the probability of the leaf nodes in $\mathcal{T}', \mathcal{T}'_k$, respectively.

Proof. Let M be the number of leaf nodes of $\mathcal{T}'$. According to the definitions of the response distribution $\mathcal{P}$ and mass distribution $\mathcal{P}_{mass}$, we have $M p^{\mathcal{D}_j} = X_{\mathcal{D}_j}, \forall j \in \{1, \dots, N\}$. Therefore,

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{P}_k} [-p^{\mathcal{D}_k} \log p^{\mathcal{D}}] &= \sum_{i=1}^M -p_i^{\mathcal{D}_k} \log p_i^{\mathcal{D}} \\ &= \sum_{i=1}^M -\frac{1}{M} X_{i, \mathcal{D}_k} \log X_{i, \mathcal{D}} + \log M \\ &= \mathbb{E}_{X_{\mathcal{D}_k} \sim \mathcal{P}_{mass}^k, X_{\mathcal{D}} \sim \mathcal{P}_{mass}} [-X_{\mathcal{D}_k} \log X_{\mathcal{D}}] + \log M. \end{aligned}$$

□

Remark A.6. In Lemma A.5, $X_{\mathcal{D}_k}$ are assumed to be independent. However due to $\sum_{i=1}^M p_i^{\mathcal{D}_k} = 1$, the $X_{\mathcal{D}_k}$ are not actually independent. Considering that M is sufficiently large in our subsequent analysis, we can regard the independence of $X_{\mathcal{D}_k}$ as a good approximation.

We assume that the mass distribution of the segment follows a heavy-tailed Pareto distribution, with supporting evidence from [Mingard et al. \(2021\)](#); [Zipf \(1946\)](#).

Assumption A.7 (The Pareto Distribution). We assume that the mass distribution of pruned token trees $\mathcal{T}$ of depth d , across different datasets, follows a Pareto distribution with identical parameters:

$$p_X(x) = \begin{cases} \frac{\alpha c^\alpha}{x^{\alpha+1}} & x \geq c, \\ 0 & x < c, \end{cases}$$

where α, c are parameters of the Pareto distribution.

Theorem A.8 (Elasticity of Language Models). *Consider the pre-training dataset $\mathcal{D}_p = \bigcup_{i=1}^3 \mathcal{D}_i$, the alignment dataset $\mathcal{D}_a$, and the perturbation dataset $\mathcal{D}_t$, with the model $p_\theta(\cdot)$ trained on $\mathcal{D} = \mathcal{D}_p \cup \mathcal{D}_a \cup \mathcal{D}_t$. Assume that $\mathcal{D}_a \stackrel{d}{\sim} \mathcal{D}_2$, $\mathcal{D}_t \stackrel{d}{\sim} \mathcal{D}_3$, and $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$ are each distributed according to a Pareto mass distribution ([Newman, 2005](#)). As the perturbation data volume $|\mathcal{D}_t|$ varies, $\gamma_{p_\theta}^{\mathcal{D}_p/\mathcal{D}}$ and $\gamma_{p_\theta}^{\mathcal{D}_a/\mathcal{D}}$ satisfy:*

$$\begin{aligned} \frac{d\gamma_{p_\theta}^{\mathcal{D}_a/\mathcal{D}}}{dl} &= \Theta \left(-k \frac{d\gamma_{p_\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl} \right), \\ \frac{d\gamma_{p_\theta}^{\mathcal{D}_p/\mathcal{D}}}{dl} &< 0, \frac{d\gamma_{p_\theta}^{\mathcal{D}_a/\mathcal{D}}}{dl} > 0, \end{aligned} \tag{4}$$

where $l = \frac{|\mathcal{D}_t|}{|\mathcal{D}_a|} \ll 1$, $k = \frac{|\mathcal{D}_p|}{|\mathcal{D}_a|} \gg 1$, and $\{\mathcal{D}_i\}_{i=1}^3$ are datasets of equal cardinality.